Notes: Covitz, Liang, and Suarez (JF, 2013)

The Evolution of a Financial Crisis: Collapse of the Asset-Backed Commercial Paper Market

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1 Big picture

- **Research question.** Did investors run on asset-backed commercial paper (ABCP) programs during the 2007 crisis, and were these runs random panics or related to observable program and macro-financial risks?
- **Core setting.** ABCP programs are bankruptcy-remote conduits that issue short-term commercial paper to fund receivables, loans, mortgage assets, securities, and structured products. By the end of 2006, U.S. ABCP outstanding was about $1.1 trillion, larger than unsecured commercial paper and central to shadow banking.
- **Main empirical fact.** ABCP outstanding fell by about $190 billion in August 2007 and by about $350 billion during the final five months of 2007. The contraction was accompanied by a jump in overnight risk spreads and a sharp shortening of new-issue maturities.
- **Run definition.** A program is in a run in a week if more than 10% of its outstanding paper is scheduled to mature but it issues no new paper; a program remains in a run until it issues again.
- **Main result on runs.** Roughly one-third of programs experienced a run within weeks of the crisis onset, and nearly 40% of programs were in a run by year-end 2007. Exiting a run was rare.
- **Risk-based selection.** Runs were more likely for programs with weak liquidity support, extendible paper, lower ratings, riskier liquidity providers, and program types more exposed to mortgage or structured-credit assets.
- **Investor pricing behavior.** For programs still able to issue, yields and maturities also varied systematically with program risk and macro-financial stress. Investors did not simply stop buying all ABCP; they repriced and shortened maturities for riskier programs.
- **Theoretical interpretation.** The evidence is consistent with asymmetric-information theories of runs: a common shock arrives, investors do not know which issuers are exposed, and they run first from observably weak institutions.
- **Shadow-bank implication.** The paper shows that runnable short-term debt in shadow banking can transmit stress to bank balance sheets through explicit and implicit liquidity support commitments.

Incremental contribution: The paper makes three incremental contributions. First, it provides one of the first program-level empirical measurements of runs in a major shadow-banking market, rather than relying only on aggregate ABCP quantities. Second, it connects the collapse of ABCP to classic banking-panic theory by showing that runs were concentrated among observably weaker programs after a common shock. Third, it studies three margins jointly—run incidence, new-issue spreads, and new-issue maturity—which lets the paper distinguish total market withdrawal from more measured risk-based repricing and maturity compression.

2 Data and measurement

2.1 ABCP market and program types

The ABCP market consisted of conduits that issued short-term paper to finance pools of assets. Program types differ sharply in assets, support, and sponsor identity:

- **Multi-seller programs** finance diversified receivables and loans from multiple sellers. These were the largest segment of the market and generally had stronger support.
- **Single-seller programs** finance assets from one originator. Mortgage single-seller programs had direct mortgage and MBS exposure.
- **Securities arbitrage programs** finance long-term highly rated securities using short-term paper.

- **Structured investment vehicles (SIVs)** and **CDOs** finance structured and mortgage-related securities and often relied on weaker or more conditional support.
- **Hybrid and other programs** include programs that do not fit the main categories.

Interpretation: The institutional distinction matters because investors in short-term paper were not necessarily observing the precise asset portfolio inside each conduit. Program type, support structure, ratings, and sponsor identity were therefore public signals about hidden portfolio exposure and rollover risk.

2.2 Transaction and outstanding data

The paper uses proprietary transaction-level and program-level data from the Depository Trust and Clearing Corporation (DTCC):

- nearly 700,000 ABCP transactions in 2007;
- 339 ABCP programs;
- transaction-level issue prices, quantities, maturities, and program identity;
- weekly program-level outstanding paper and maturity structure.

These data allow the authors to measure not just aggregate market contraction but also whether each program could roll maturing paper, what yield it paid when issuing, and how short its new issues became.

Interpretation: The main advantage of the data is granularity. Aggregate ABCP outstanding can show a market collapse, but only program-level issuance and maturities can identify which programs were unable to roll debt and which still issued at worse terms.

2.3 Program characteristics

The authors supplement DTCC data with hand-collected information from Moody's reports:

- **Extendibility:** equals one if the issuer can extend paper maturity. This is a weak form of liquidity support because investors bear more rollover risk.
- **Number of liquidity providers:** proxies for breadth and strength of liquidity support.
- **Lower rating:** equals one if the program is rated below Moody's P-1.
- **Credit support:** equals one if financial institutions commit to support the program in the event of asset impairment.
- **Initial average maturity:** average maturity of outstanding paper at the start of the sample window.
- **CDS spread of main liquidity provider:** time-varying measure of the perceived risk of the financial institution most important to the conduit support arrangement.

Interpretation: These variables are designed to capture the observable weakness of the program. They are not perfect measures of actual subprime exposure, but they proxy for what commercial paper investors could see when deciding whether to roll.

2.4 Program type, sponsor type, and macro-financial variables

Program type indicators include multi-seller, nonmortgage single-seller, mortgage single-seller, securities arbitrage, SIV, CDO, and other. Sponsor type indicators include large U.S. banks, small U.S. banks, foreign banks, and nonbanking sponsors.

Macro-financial variables include:

- the one-month LIBOR-OIS spread;
- the volatility of the one-month LIBOR-OIS spread;
- the return on the ABX index;
- the volatility of the ABX return.

Interpretation: LIBOR-OIS captures funding stress and perceived interbank risk. ABX returns capture perceived mortgage-credit deterioration. Volatility measures capture uncertainty, which matters because runs often occur when investors cannot determine exactly which issuers are exposed.

2.5 Outcome variables

The main outcomes are:

- **Run indicator:** whether the program fails to issue in a week when more than 10% of outstanding paper matures, or remains unable to issue after an earlier run.
- **Overnight risk spread:** weighted average spread on overnight ABCP over the target federal funds rate.
- **Average maturity of new issues:** weighted average days to maturity of newly issued paper.

Interpretation: The three outcomes map to three investor responses: refusing to roll at any price, demanding higher compensation, and demanding shorter maturity. Studying them jointly is essential because some programs exited through outright runs while others survived by issuing shorter, more expensive paper.

3 Models and empirical specifications

3.1 Run definition

The run indicator is defined as:

$$Run_{it} = \begin{cases} 1, & \text{if } Maturing_{it}/Outstanding_{it} > 0.1 \text{ and } Issuance_{it} = 0, \\ 1, & \text{if } Run_{i,t-1} = 1 \text{ and } Issuance_{it} = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Interpretation: The 10% threshold captures a meaningful rollover need. The zero-issuance condition is conservative: a program that issued a tiny amount at a high spread is not classified as in a run. The definition also cannot detect whether paper was bought by a related sponsor rather than by arm's-length investors.

3.2 Cross-sectional probit for experiencing a run

For the pre-crisis and crisis periods separately, the authors estimate:

$$Pr(Run_i = 1) = \Phi\left(\alpha + \sum_h \beta_h ProgramChar_{hi} + \sum_j \eta_j ProgramType_{ji} + \sum_k \gamma_k SponsorType_{ki}\right). \tag{2}$$

Interpretation: The specification asks whether programs with weaker ex ante observable characteristics were more likely to experience at least one run during the period. Estimating separately before and after August 2007 tests whether the same characteristics became more important after the common shock.

3.3 Weekly panel probit for runs

The panel model is:

$$Pr(Run_{it} = 1) = \Phi\left(\alpha + \sum_g \beta_g ProgramChar'_{git} + \sum_j \eta_j ProgramType_{ji} + \sum_k \gamma_k SponsorType_{ki} + \sum_l \delta_l Macro_{lt}\right). \tag{3}$$

Interpretation: This adds time-varying ratings, liquidity-provider CDS spreads, and macro-financial stress. The panel model therefore tests whether runs were more likely when both program-specific weakness and aggregate uncertainty were high.

3.4 Spread and maturity regressions

For programs that issue, the paper estimates panel regressions for:

$$Spread_{it} = \alpha + \beta' ProgramChar_{it} + \eta' ProgramType_i + \gamma' SponsorType_i + \delta' Macro_t + u_{it}, \tag{4}$$

$$Maturity_{it} = \alpha + \beta' ProgramChar_{it} + \eta' ProgramType_i + \gamma' SponsorType_i + \delta' Macro_t + u_{it}. \tag{5}$$

Interpretation: The spread regressions test whether investors priced risk among programs that could still borrow. The maturity regressions test whether investors demanded shorter rollover intervals for risky programs, consistent with dynamic coordination theories of short-term debt.

4 Identification strategy and interpretation

4.1 Empirical strategy

The paper is not a randomized experiment. It is an empirical diagnostic of a financial panic using detailed program-level data. The strategy has five steps:

1. Define a conservative program-level run measure from maturity needs and issuance failure.
2. Split 2007 into a pre-crisis period, February-July, and a crisis period, August-December.
3. Show that runs rise abruptly after the common shock to mortgage and funding markets.
4. Test whether runs are related to observable program weakness rather than being random across programs.
5. Test whether programs that still issue face higher spreads and shorter maturities based on the same risk characteristics.

4.2 Why the design is informative

The key identifying logic is that asymmetric-information run theory predicts a common shock with uncertain incidence across issuers. If the ABCP collapse were a purely indiscriminate panic, observable program characteristics should not explain which programs ran, paid higher spreads, or shortened maturities. If investors inferred risk from public signals, weaker program structures should predict all three outcomes.

4.3 Main assumptions

- Observable program characteristics proxy for the information investors used to assess rollover and asset risk.
- DTCC issuance data accurately distinguish programs that could and could not roll paper.
- The pre-crisis/crisis split captures the change in investor beliefs after the subprime and money-market shocks.
- Macro variables capture aggregate funding stress and mortgage-credit stress that affect all programs.

4.4 Threats and limitations

- The data do not reveal investor identities, so the authors cannot observe whether the same investors ran across programs or whether sponsors bought paper to prevent a visible run.
- The run definition is conservative and may miss partial runs or non-arm's-length rollovers.
- Program characteristics are proxies, not direct measures of hidden asset exposure.
- The analysis documents systematic selection and pricing, but it does not claim a causal treatment effect of a single program characteristic.

4.5 Identification takeaway

The empirical design is strongest as evidence against random panic and in favor of information-based run selection. It shows that the crisis was aggregate in origin but program-specific in incidence.

5 Tables and figures

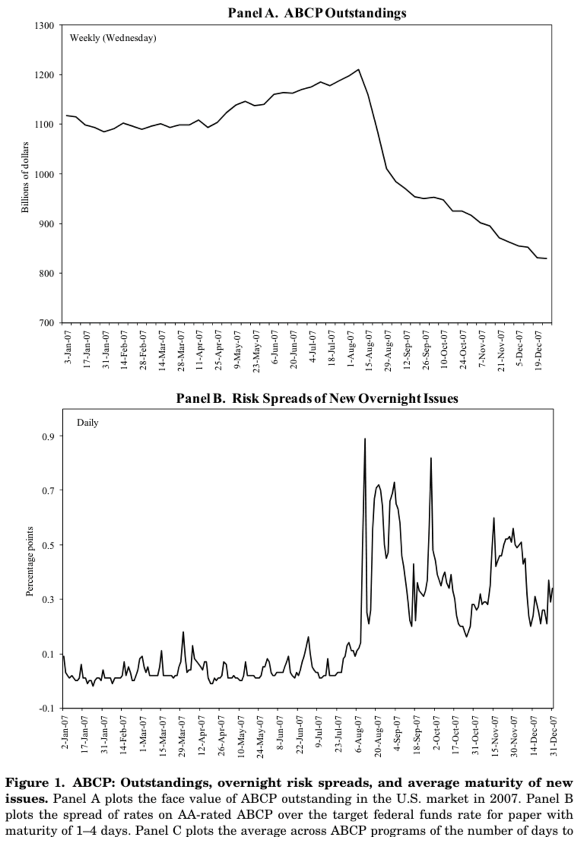
5.1 Figure 1: ABCP outstanding, overnight risk spreads, and average maturity of new issues

Read:

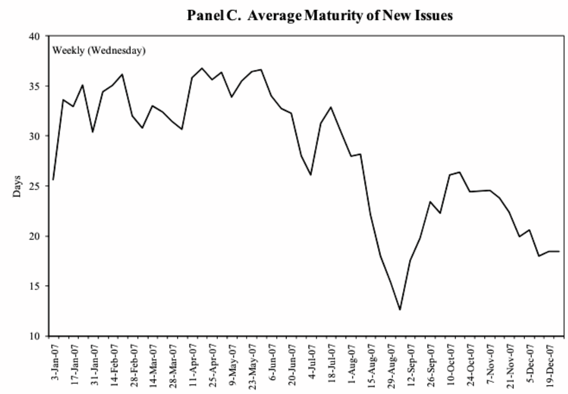
- Panel A shows total ABCP outstanding rising through July 2007 and then dropping sharply in August and continuing to fall through year-end.
- Panel B shows overnight spreads jumping from near zero to crisis levels after early August.
- Panel C shows new-issue maturities falling sharply, from roughly 30-35 days before the crisis to much shorter maturities in August and September.

Interpretation: The three panels together show a collapse in quantity, a rise in price of funding, and a contraction in maturity. The ABCP market did not merely shrink; its entire short-term funding technology changed.

Economic message: Figure 1 motivates the central question: was the collapse a broad liquidity shock, a risk-based run, or both?



(a) Outstanding and overnight spreads



(b) Average maturity of new issues

Figure 1. ABCP: Outstanding, overnight risk spreads, and average maturity of new issues

5.2 Table I: ABCP program types

Read:

- Multi-seller programs account for 45% of outstanding ABCP and 98 of the 339 programs.
- Nonmortgage single-seller, mortgage single-seller, securities arbitrage, SIV, CDO, and hybrid programs differ strongly in asset type, sponsor type, extendibility, and support.
- Mortgage single-seller, SIV, and CDO programs are more likely to be tied to mortgage or structured assets.

Interpretation: Table I shows why program type can be a meaningful public signal. Investors could use program type to infer potential mortgage exposure and support quality even when underlying asset details were opaque.

Economic message: The ABCP market was not homogeneous. Some structures resembled safer receivables conduits; others resembled maturity-transformation vehicles exposed to structured finance.

Table I

ABCP Program Types

This table describes ABCP programs by program type, as of July 2007 (when ABCP outstandings peaked). Data on outstandings are from DTCC and program characteristics are collected from reports by Moody's.

Program Type	Assets	Number of Programs	Market Share of Outstandings (%)	Programs with Extendible Paper (%)	Programs with Credit Support (%)	Programs with Large US Bank Sponsors (%)	Programs with Small US Bank Sponsors (%)	Programs with Foreign Bank Sponsors (%)	Programs with Nonbank Sponsors (%)
Multi-seller	Diversified receivables and loans	98	45	19	30	19	4	57	20
Nonmortgage single seller	Credit-card receivables and auto loans	40	11	62	28	8	10	8	74
Mortgage single seller	Mortgages and mortgage-backed securities (MBS)	11	2	67	0	9	9	0	82
Securities arbitrage	Highly rated long-term securities, including MBS	35	13	9	17	14	11	66	9
Structured investment vehicle	Highly rated long-term securities, including MBS	35	7	0	0	11	0	23	66
CDO	Highly rated long-term securities, including MBS	36	4	25	0	0	3	3	94
Hybrid and other	N.A.	84	18	20	10	5	2	12	81
Total		339	100	24	16	10	5	30	55

5.3 Table II: ABCP outstandings, overnight risk spreads, and average maturity of new issues by program type

Read:

- Total outstanding falls from $1.163 trillion in July to $816 billion by December 2007.
- Multi-seller programs decline relatively little compared with single-seller mortgage programs, SIVs, CDOs, and securities arbitrage programs.
- Overnight spreads rise sharply in August and September, with especially large increases for riskier program types.
- Average maturity falls from about 30 days in July to 22 days in August and 18 days by December, with the sharpest maturity compression in risky categories.

Interpretation: Table II documents heterogeneity behind the aggregate collapse. Program types likely exposed to mortgage assets and weaker support lost more funding and faced worse terms.

Economic message: The crisis was not simply a market-wide freeze. Investors continued to differentiate across conduit types.

5.4 Figure 2: Runs on ABCP programs

Read:

- The fraction of programs in run status rises sharply in August and reaches roughly 40% by year-end.
- The hazard of leaving a run state falls sharply after the crisis begins and approaches very low levels by late 2007.
- Early in the year, the few run episodes appear more transitory; after August, runs become persistent.

Interpretation: Figure 2 turns the aggregate ABCP contraction into a program-level run statistic. The persistence of the run state suggests that many programs were effectively shut out of the market.

Economic message: This figure is the core evidence that the ABCP collapse included actual creditor runs, not just lower voluntary issuance.

Table II

ABCP: Outstandings, Overnight Risk Spreads, and Average Maturity of New Issues, by Program Type

Panel A reports the amount of paper outstanding at the end of each month in 2007 for all program types in the U.S. ABCP market. Panel B reports the spread of rates on overnight ABCP issues, by program type, over the target federal funds rate. Spreads are weighted averages of spreads on individual transactions using face value of transactions as weights. Panel C reports the average number of days to maturity of newly issued paper. Data are from DTCC and program type classification is from Moody's.

Panel A: Outstandings

Billions of Dollars, End of the Month		Total	Multi-Seller	Non-mortgage Single Seller	Mortgage Single Seller	Securities Arbitrage	Structured Investment Vehicle	CDO	Hybrid and Other
2007	January	1,061	455	121	32	159	63	41	190
	February	1,067	459	129	33	154	60	41	190
	March	1,070	480	122	25	148	56	46	193
	April	1,092	492	125	32	142	63	46	193
	May	1,125	503	126	35	149	65	46	202
	June	1,151	518	123	23	150	79	48	211
	July	1,163	525	126	23	148	84	47	210
	August	976	503	79	4	120	70	39	160
	September	927	484	74	2	133	49	33	153
	October	896	465	68	2	140	29	32	160
	November	838	461	55	1	117	22	31	152
	December	816	469	51	2	102	15	27	151

Panel B: Overnight Risk Spreads

Percentage Points Month Average		Total	Multi-Seller	Non-mortgage Single Seller	Mortgage Single Seller	Securities Arbitrage	Structured Investment Vehicle	CDO	Hybrid and Other
2007	January	0.02	0.02	0.00	0.05	0.02	0.01	0.02	0.02
	February	0.02	0.02	0.01	0.04	0.03	0.01	0.03	0.03
	March	0.05	0.05	0.06	0.07	0.04	0.04	0.10	0.04
	April	0.05	0.05	0.05	0.06	0.04	0.04	0.09	0.04
	May	0.03	0.03	0.03	0.06	0.03	0.02	0.04	0.03
	June	0.06	0.06	0.07	0.09	0.06	0.05	0.07	0.05
	July	0.06	0.06	0.05	0.08	0.05	0.05	0.07	0.05
	August	0.47	0.44	0.42	0.76	0.47	0.44	0.51	0.55
	September	0.49	0.41	0.71	1.22	0.53	0.55	0.41	0.65
	October	0.34	0.24	0.83	1.51	0.42	0.55	0.50	0.47
	November	0.44	0.35	1.01	1.75	0.57	0.76	0.54	0.50
	December	0.53	0.41	0.91	1.92	0.69	1.11	0.75	0.53

(*Continued*)

Table II. ABCP: Outstandings, overnight risk spreads, and average maturity of new issues by program type

Table II—*Continued*

Panel C: Average Maturity of New Issues

Days to Maturity, Month Average		Total	Multi-Seller	Non-mortgage Single Seller	Mortgage Single Seller	Securities Arbitrage	Structured Investment Vehicle	CDO	Hybrid and Other
2007	January	33	25	36	16	37	47	45	35
	February	34	26	38	17	44	55	41	32
	March	32	24	35	17	35	43	42	34
	April	35	26	37	18	45	57	42	32
	May	35	26	40	16	49	56	40	32
	June	32	23	35	14	37	41	45	34
	July	30	23	36	16	36	42	34	30
	August	22	18	25	7	24	45	24	19
	September	19	16	21	7	18	25	27	18
	October	24	17	29	3	35	47	29	24
	November	22	20	22	2	25	22	28	22
	December	18	16	15	2	18	25	31	18

Table II. ABCP: Outstandings, overnight risk spreads, and average maturity of new issues by program type

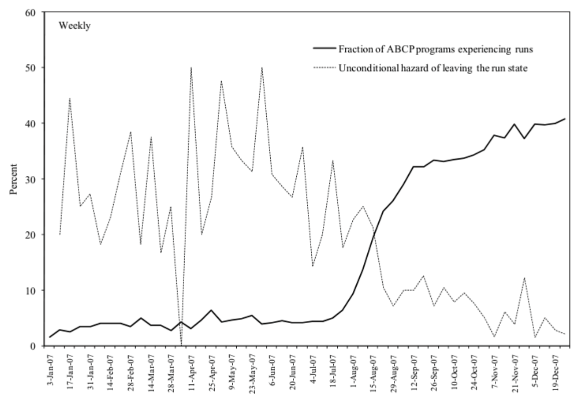


Figure 2. Runs on ABCP programs. The solid line plots the percent of programs experiencing a run. We define a program as experiencing a run in weeks when it does not issue paper but has at least 10% of paper maturing or when the program continues to not issue paper after experiencing a run in the previous week (see equation (1) in the text). The dotted line plots the unconditional probability of not experiencing a run in a given week after having experienced a run in the previous week (i.e., the hazard rate of exiting a run). The figure is based on weekly data from DTCC on paper outstanding, maturities, and issuance for 339 ABCP programs in 2007.

Table III
Calendar of Events in Money Markets in 2007

Month	Events in Money Markets
July	• Countrywide's disappointing earnings announcement (July 24)
August	• American Home Mortgage declares bankruptcy (Aug 6)
	• Three single-seller mortgage ABCP programs extend the maturity of their paper (Aug 6)
	• BNP halts redemptions at three affiliated funds (Aug 9)
	• ECB injects liquidity in money markets (Aug 9)
	• Federal Reserve provides liquidity (Aug 10)
	• Canadian ABCP market seizes up (Aug 14)
	• Countrywide taps on its credit lines (Aug 16)
	• Federal Reserve cuts primary credit rate 50 basis points (Aug 17)
	• An ABCP program affiliated with KKR Financial extends the maturity of its paper (Aug 20)
	• Two SIV programs default on their ABCP (Aug 22–23)
	• A second ABCP program affiliated with KKR Financial extends the maturity of its paper (Aug 23)
	• Clarification that investment-quality ABCP is accepted as discount-window collateral at the Federal Reserve (Aug 24)
September	• An SIV program sponsored by Cheyne Capital Management draws on its credit lines (Aug 30)
	• Moody's downgrades or places under review the ratings of several ABCP programs issued by SIVs (Sept 5)
	• SIFMA, the American Securitization Forum, and the European Securitization Forum recommend disclosure of holdings by ABCP programs (Sept 12)
	• Federal Reserve cuts the target federal funds target rate by 50 basis points (Sept 18)
October	• Citigroup, Bank of America, and JP Morgan Chase propose the M-LEC to backstop paper issued by SIVs (Oct 15)
	• An SIV program sponsored by Cheyne Capital Management defaults (Oct 17)
	• An SIV program sponsored by IKB Credit Management defaults (Oct 18)
	• Federal Reserve cuts the target federal funds rate by 25 basis points (Oct 31)
November	• Moody's Investors Service downgrades and places under review several SIVs (Nov 7)
December	• S&P downgrades many SIVs (Dec 7)
	• Federal Reserve cuts the target federal funds rate by 25 basis points (Dec 11)
	• Federal Reserve establishes the Term Auction Facility (TAF) and coordinates foreign exchange swap lines with other major central banks (Dec 12)
	• Citigroup announces that it will support its own-sponsored SIVs (Dec 13)
	• First TAF auction (Dec 17)
	• Citigroup, Bank of America, and JP Morgan Chase abandon the idea of M-LEC (Dec 21)

It is worth noting, however, that, while concerns about subprime mortgages appeared to precipitate turmoil in the ABCP market, the evolution of such turmoil through August and over the remainder of the year seemed not to

5.5 Table III: Calendar of events in money markets in 2007

Read:

- The table lists the sequence from Countrywide's July earnings shock, American Home Mortgage bankruptcy, BNP fund redemption halt, ECB and Federal Reserve liquidity actions, Canadian ABCP market freeze, SIV downgrades and defaults, and the eventual Term Auction Facility.
- The timing highlights that ABCP runs intensified amid both mortgage-credit news and funding-market interventions.

Interpretation: Table III connects the program-level analysis to the broader 2007 funding-market crisis. It shows that the ABCP run occurred during an evolving sequence of information shocks and policy responses.

Economic message: The run was not caused by a single isolated event; it reflected accumulating concern about mortgage exposure, sponsor support, and short-term funding market fragility.

5.6 Figure 3: Measures of financial market risk

Read:

- Panel A shows the LIBOR-OIS spread and its volatility rising sharply after August.
- Panel B shows ABX returns becoming volatile, capturing turmoil in mortgage-credit risk perceptions.
- LIBOR-OIS volatility remains elevated even when mortgage-specific measures fluctuate, suggesting broad funding uncertainty.

Interpretation: Figure 3 motivates the macro variables in the regressions. The paper tests whether program-level runs, spreads, and maturities respond to aggregate funding and mortgage-risk conditions.

Economic message: The crisis combines a mortgage-information shock with a money-market funding shock; both affect ABCP rollover decisions.

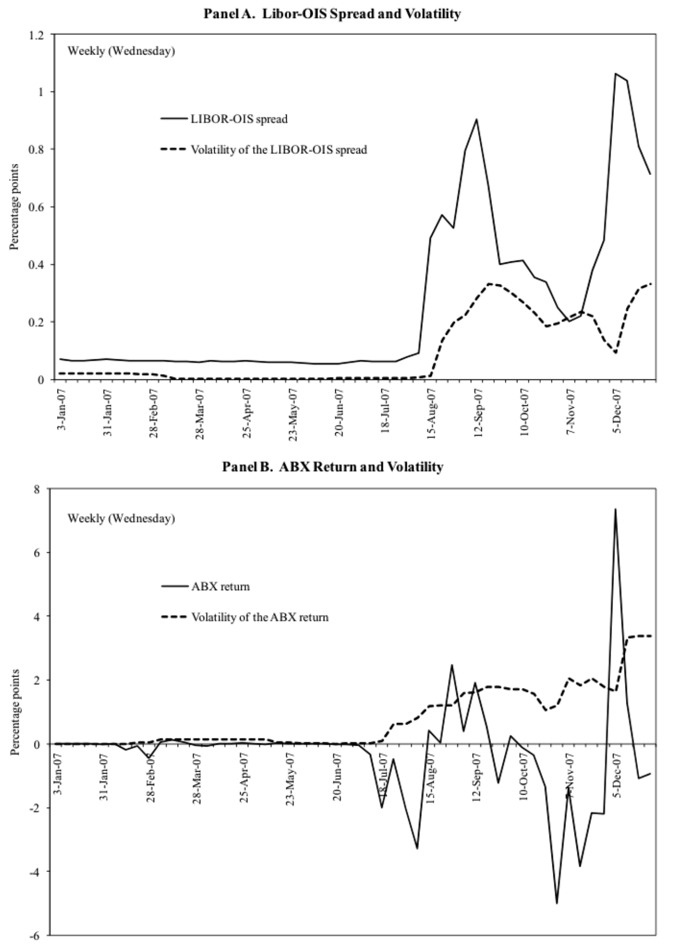


Figure 3. Measures of financial market risk. Panel A plots the weekly average spread of 1-month U.S. LIBOR over comparable maturity OIS and its volatility (Source: British Bankers

5.7 Table IV: Cross-sectional regressions of the probability of experiencing a run

Read:

- In the crisis period, extendibility raises the probability of experiencing a run; more liquidity providers reduce it; lower rating increases it.
- Multi-seller programs are less likely to run, while mortgage single-seller programs and some structured program types are more vulnerable.
- Small U.S. bank sponsorship is associated with greater run risk relative to large U.S. bank sponsorship.

Interpretation: The cross-sectional results show that runs were not random. Investors appear to have used observable indicators of weak liquidity support, lower credit quality, and program structure to decide whether to roll.

Economic message: This table is key evidence for asymmetric-information run theory: the aggregate shock arrived broadly, but investors ran from programs whose observable characteristics suggested higher hidden exposure.

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Table IV

Cross-Sectional Regressions of the Probability of Experie

This table reports the results of estimating the probit model in equation (2) usin
of ABCP programs:

$$\Pr(Run_i = 1) = F\Bigg(\alpha + \sum_h \beta_h Program\ characteristics_{hi} + \sum_j \eta_j Program$$

$$+ \sum_k \gamma_k Sponsor\ type_{ki}\Bigg).$$

The dependent variable is the probability of experiencing a run as defined in eq
point over the sample (February to July 2007 in column (1) and August to Decembe
2). F denotes the cumulative distribution function of a standard normal random va
characteristics include: an indicator for extendibility (which equals one for prog
the option to extend the maturity of their paper at the issuer's request), the nu
providers, an indicator for initial lower rating (which equals one for programs
by Moody's 1 month before the sample), an indicator for credit support (which
sponsoring financial institutions commit to support the program in the event of as
and the initial average maturity of commercial paper outstanding. *Program typ*
program i is type j and equals zero otherwise. The set of j program types inclu
nonmortgage single-seller conduits, mortgage single-seller conduits, securities arb
SIVs, CDOs, and other programs (the omitted category). *Sponsor type*$_{ki}$ equals on
sponsored by an institution of type k and equals zero otherwise. The set of k sponso
U.S. banks (the omitted category), small U.S. banks, non-U.S. banks, and nonbank
Perfect predictors of success or failure are dropped from the regression. The table r
marginal effects, and robust standard errors are reported in parentheses. *** ind
significance at the 1% level, ** at the 5% level, and * at the 10% level.

		February–July 2007 (1)
Program characteristics	Extendibility	0.185*
		(0.095)
	Number of liquidity providers	− 0.009*
		(0.005)
	Initial lower rating	dropped
		(perf. pred.)
	Credit support	0.132
		(0.090)
	Initial average maturity of outstandings	− 0.002
		(0.002)
Program type variables	Multi-seller	− 0.143**

Table IV. Cross-sectional regressions of the probability of experiencing a run

Table IV—*Continued*

		February–July 2007 (1)	August–December 2007 (2)
Sponsor type variables	Small U.S. bank sponsor	− 0.085	0.213*
		(0.097)	(0.124)
	Non-U.S. bank sponsor	− 0.162***	0.126
		(0.062)	(0.104)
	Nonbanking sponsor	− 0.118	0.108
		(0.083)	(0.104)
	Observations	240	245
	Pseudo-R^2	0.134	0.255

suggesting fewer concerns about diversified conduits with little or no mortgage holdings; the corresponding effect in the pre-crisis period was significant at the 5% level and notably smaller in absolute magnitude. The marginal effect of *Mortgage single-seller* is positive though insignificant in the crisis period, and

Table IV. Cross-sectional regressions of the probability of experiencing a run

5.8 Table V: Panel regressions of the probability of experiencing a run

Read:

- In the crisis period, extendibility and lower rating strongly predict weekly run probability.
- The CDS spread of the main liquidity provider is positively related to run probability, meaning investors cared about the sponsor's ability to support the conduit.
- Volatility of the LIBOR-OIS spread is positively related to runs, while pre-crisis macro coefficients are weak.

Interpretation: The panel results strengthen the cross-sectional evidence by adding time variation in support risk and aggregate market risk. A program with a riskier support provider was more vulnerable precisely when funding-market uncertainty rose.

Economic message: The conduit run was linked to bank balance-sheet risk: investors judged ABCP programs partly by the perceived strength of the banks standing behind them.

Table V

Panel Regressions of the Probability of Experiencing a Run

This table reports the results of estimating the probit model in equation (3) from the text using panels of weekly observations of ABCP programs from February to July 2007 (columns 1 and 2) and from August to December 2007 (columns 3 and 4):

$$\Pr(Run_{it}=1) = F\left(\alpha + \sum_g \beta_g Program\ characteristics_{g'it} + \sum_j \eta_j Program\ type_{ji} + \sum_k \gamma_k Sponsor\ type_{ki} + \sum_l \delta_l Macro\ variables_{lt}\right).$$

The dependent variable is the probability of experiencing a run during week t as defined in equation (1). F denotes the cumulative distribution function of a standard normal random variable. *Program characteristics* include: an indicator for extendibility (which equals one for programs that have the option to extend the maturity of their paper at the issuer's request); the number of liquidity providers; the 5-year CDS spread on week t for the institution listed as the main liquidity provider for program i; a time-varying indicator for lower rating (which equals one for programs rated below P-1 by Moody's in week t); an indicator for credit support (which equals one when sponsoring financial institutions commit to support the program in the event of asset impairment); and the initial average maturity of commercial paper outstanding. *Program type*$_{ji}$ and *Sponsor type*$_{ki}$ are defined as in Table IV. In columns 1 and 3, the *Macro variables* are the weekly average *Spread of one-month LIBOR over OIS* and its volatility. In columns 2 and 4, the *Macro variables* are the weekly *Return on the ABX* index and its volatility. Perfect predictors of success or failure are dropped from the regression. The table reports estimated marginal effects, and standard errors clustered by program are reported in parentheses. *** indicates statistical significance at the 1% level, ** at the 5% level, and * at the 10% level.

		February–July 2007 (1)	February–July 2007 (2)	August–December 2007 (3)	August–December 2007 (4)
Program characteristics	Extendibility	− 0.010	− 0.009	0.462***	0.467***
		(0.028)	(0.028)	(0.116)	(0.116)
	Number of liquidity providers	− 0.022**	− 0.022**	− 0.008	− 0.008
		(0.010)	(0.010)	(0.007)	(0.007)
	CDS spread of main liquidity provider	0.236*	0.273	0.359***	0.277**
		(0.131)	(0.167)	(0.119)	(0.117)
	Lower rating	dropped	dropped	0.345***	0.345***
		(perf. pred.)	(perf. pred.)	(0.118)	(0.121)
	Credit support	0.010	0.009	0.092	0.094
		(0.030)	(0.029)	(0.121)	(0.122)
	Initial average maturity of outstandings	− 0.001	− 0.001	0.001	0.001
		(0.001)	(0.001)	(0.002)	(0.002)
Program type variables	Multi-seller	− 0.056*	− 0.055*	− 0.239***	− 0.240***
		(0.029)	(0.028)	(0.072)	(0.072)
	Nonmortgage single	− 0.017	− 0.017	− 0.060	− 0.064

Table V. Panel regressions of the probability of experiencing a run

Table V—*Continued*

		February–July 2007 (1)	February–July 2007 (2)	August–December 2007 (3)	August–December 2007 (4)
	Structured investment vehicle	dropped	dropped	0.302***	0.314***
		(perf. pred.)	(perf. pred.)	(0.116)	(0.114)
	CDO	− 0.025**	− 0.025**	− 0.043	− 0.031
		(0.011)	(0.011)	(0.161)	(0.167)
Sponsor type variables	Small U.S. bank sponsor	− 0.031**	− 0.031**	0.382**	0.384**
		(0.013)	(0.013)	(0.159)	(0.160)
	Non-U.S. bank sponsor	− 0.036*	− 0.035*	0.127	0.119
		(0.020)	(0.021)	(0.110)	(0.110)
	Nonbanking sponsor	− 0.024	− 0.022	0.072	0.062
		(0.028)	(0.028)	(0.088)	(0.089)
Macro variables	Spread of 1-month LIBOR over OIS	− 0.117		0.040	
		(0.815)		(0.028)	
	Volatility of the spread of one-month	0.136		0.582***	
		(0.558)		(0.127)	

Table V. Panel regressions of the probability of experiencing a run

5.9 Table VI: Panel regressions of risk spreads on overnight ABCP issues

Read:

- During the crisis, extendible programs, programs with fewer liquidity providers, lower-rated programs, and programs with riskier liquidity providers pay higher spreads.
- Program type matters more in the crisis than before, especially for mortgage and structured-credit-related programs.
- LIBOR-OIS volatility is associated with higher spreads, reinforcing the role of aggregate funding stress.

Interpretation: Table VI shows that even among programs that could still issue, investors demanded compensation for observable risk. Runs were the extreme outcome, but repricing occurred along the intensive margin.

Economic message: ABCP investors were risk intolerant but not completely indiscriminate. They priced risk when the program was still marketable and refused to roll when risk or uncertainty was too high.

Table VI
Panel Regressions of Risk Spreads on Overnight ABCP Issues

This table reports the results of estimating the following equation using panels of daily observations of ABCP programs from February to July 2007 (columns 1 and 2) and from August to December 2007 (columns 3 and 4):

$$Spread_{it} = \alpha + \sum_g \beta_g Program\ characteristics_{g'it} + \sum_j \eta_j\ Program\ type_{ji} + \sum_k \gamma_k Sponsor\ type_{ki} + \sum_l \delta_l Macro\ variables_{lt} + \varepsilon_{it}.$$

The dependent variable, $Spread_{it}$, is the spread over the target federal funds rate paid by program i on day t to issue overnight paper (1–4 days of maturity). *Program characteristics* include: an indicator for extendibility (which equals one for programs that have the option to extend the maturity of their paper at the issuer's request); the number of liquidity providers; the 5-year CDS spread on week t for the main liquidity provider for program i; a time-varying indicator for lower rating (which equals one for programs rated below P-1 by Moody's in week t); an indicator for credit support (which equals one when sponsoring financial institutions commit to support the program in the event of asset impairment); and the initial average maturity of commercial paper outstanding. $Program\ type_{ji}$ and $Sponsor\ type_{ki}$ are defined as in Table IV. In columns 1 and 3, the *Macro variables* are the weekly average *Spread of one-month LIBOR over OIS* and its volatility. In columns 2 and 4, the Macro variables are the weekly *Return on the ABX index* and its volatility. Standard errors clustered by program are reported in parentheses. *** indicates statistical significance at the 1% level, ** at the 5% level, and * at the 10% level.

		February–July 2007 (1)	February–July 2007 (2)	August–December 2007 (3)	August–December 2007 (4)
Program characteristics	Extendibility	0.032***	0.032***	0.393**	0.396**
		(0.009)	(0.009)	(0.160)	(0.162)
	Number of liquidity providers	−0.001	−0.001	−0.010**	−0.010**
		(0.001)	(0.001)	(0.004)	(0.004)
	CDS spread of main liquidity provider	0.073***	0.090***	0.291***	0.374***
		(0.016)	(0.020)	(0.096)	(0.118)
	Lower rating	0.076***	0.076***	0.408***	0.416***
		(0.009)	(0.009)	(0.145)	(0.141)
	Credit support	0.001	0.002	−0.110	−0.110
		(0.006)	(0.006)	(0.073)	(0.072)
	Initial average maturity of outstandings	0.000	0.000	0.000	0.000
		(0.000)	(0.000)	(0.002)	(0.002)
Program type variables	Multi-seller	−0.005	−0.005	−0.138**	−0.135**
		(0.012)	(0.012)	(0.064)	(0.063)
	Nonmortgage single seller	−0.006	−0.005	0.175	0.185
		(0.023)	(0.023)	(0.155)	(0.155)
	Mortgage single seller	0.005	0.004	0.564**	0.548**
		(0.014)	(0.014)	(0.240)	(0.254)
	Securities arbitrage	−0.013	−0.014	−0.013	−0.011
		(0.012)	(0.012)	(0.098)	(0.097)
	Structured investment vehicle	0.007	0.007	0.055	0.045
		(0.013)	(0.013)	(0.135)	(0.137)

(Continued)

Table VI. Panel regressions of risk spreads on overnight ABCP issues

Table VI—*Continued*

		February–July 2007 (1)	February–July 2007 (2)	August–December 2007 (3)	August–December 2007 (4)
	CDO	−0.020	−0.022	(dropped)	(dropped)
		(0.014)	(0.013)		
Sponsor type variables	Small U.S. bank sponsor	0.060***	0.060***	0.146	0.130
		(0.010)	(0.009)	(0.182)	(0.179)
	Non-U.S. bank sponsor	0.018***	0.019***	0.123	0.132
		(0.005)	(0.006)	(0.083)	(0.083)
	Nonbanking sponsor	0.025***	0.027***	0.164**	0.180***
		(0.005)	(0.005)	(0.063)	(0.064)
Macro variables	Spread of one-month LIBOR over OIS	−0.563		0.172***	
		(0.554)		(0.038)	
	Volatility of the spread of one-month LIBOR over OIS	12.819***		0.208**	
		(0.988)		(0.091)	
	Return on the ABX index		−0.005***		0.045***
			(0.001)		(0.004)
	Volatility of the return on the ABX index		0.004		−0.019
			(0.005)		(0.025)
	Constant	0.015	0.012	0.148	0.227**

Table VI. Panel regressions of risk spreads on overnight ABCP issues

5.10 Table VII: Panel regressions of average maturity of new issues

Read:

- In the crisis period, extendibility and lower ratings predict shorter new-issue maturities.

- Stronger support variables, such as more liquidity providers and credit support, are associated with longer maturities.
- Macro-financial uncertainty, especially LIBOR-OIS stress and volatility, is linked to shorter maturities.

Interpretation: Table VII shows that investors shortened maturities for riskier programs and during periods of funding stress. This is consistent with models in which short-term creditors prefer more frequent rollover dates when asset values are uncertain.

Economic message: Maturity shortening was a form of market discipline and fragility. It gave investors more frequent exit opportunities while increasing rollover pressure on ABCP programs.

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Table VII

Panel Regressions of Average Maturity of New Issues

This table reports the results of estimating the following equation using panels of weekly observations from February to July 2007 (columns 1 and 2) and from August to December 2007 (columns 3 and 4):

$$\mathit{Average\ maturity}_{it} = \alpha + \sum_g \beta_g \mathit{Program\ characteristics}_{git} + \sum_j \eta_j\ \mathit{Program\ type}_{ji} + \sum_k \gamma_k \mathit{Sponsor\ type}_{ki} + \sum_l \delta_l \mathit{Macro\ variables}_{lt} + \varepsilon_{it}.$$

The dependent variable, *Average maturity*$_{it}$, is the average maturity (in days) of new issues of ABCP for program i in week t. *Program characteristics* include: an indicator for extendibility (which equals one for programs that have the option to extend the maturity of their paper at the issuer's request), the number of liquidity providers, the 5-year CDS spread on week t for the institution listed as the main liquidity provider for program i, a time-varying indicator for lower rating (which equals one for programs rated below P-1 by Moody's in week t), an indicator for credit support (which equals one when sponsoring financial institutions commit to support the program in the event of asset impairment), and the initial average maturity of commercial paper outstanding. *Program type*$_{ji}$ and *Sponsor type*$_{ki}$ are defined as in Table IV. In columns 1 and 3, the *Macro variables* are the weekly average *Spread of one-month LIBOR over OIS* and its volatility. In columns 2 and 4, the *Macro variables* are the weekly *Return on the ABX index* and its volatility. Standard errors clustered by program are reported in parentheses. *** indicates statistical significance at the 1% level, ** at the 5% level, and * at the 10% level.

		February–July 2007 (1)	February–July 2007 (2)	August–December 2007 (3)	August–December 2007 (4)
Program characteristics	Extendibility	7.225	7.201	−6.327**	−6.115*
		(6.299)	(6.341)	(3.055)	(3.199)
	Number of liquidity providers	0.322	0.338	0.731***	0.731***
		(0.217)	(0.223)	(0.164)	(0.166)
	CDS spread of main liquidity provider	−11.275	−19.041	−5.332	−5.249
		(24.212)	(28.464)	(8.783)	(9.091)
	Lower rating	2.965	2.692	−12.018***	−11.794***
		(3.157)	(3.160)	(3.235)	(3.179)
	Credit support	−1.143	−1.079	11.599**	11.772**
		(4.689)	(4.712)	(5.519)	(5.549)
	Initial average maturity of outstandings	1.027***	1.024***	0.357**	0.356**
		(0.162)	(0.161)	(0.159)	(0.161)
Program type variables	Multi-seller	−2.475	−2.541	−7.978	−8.297
		(3.669)	(3.665)	(5.467)	(5.470)
	Nonmortgage single seller	1.896	1.827	−16.179**	−16.500**
		(6.893)	(6.898)	(7.401)	(7.416)
	Mortgage single seller	−9.477	−9.084	−17.533***	−16.938***
		(10.294)	(10.270)	(5.315)	(5.280)
	Securities arbitrage	4.379	4.523	−4.313	−4.787
		(6.901)	(6.900)	(8.269)	(8.290)
	Structured investment vehicle	2.359	2.400	−7.132	−6.757
		(6.407)	(6.434)	(10.968)	(11.155)

(Continued)

Table VII. Panel regressions of average maturity of new issues

Table VII—*Continued*

		February–July 2007 (1)	February–July 2007 (2)	August–December 2007 (3)	August–December 2007 (4)
	CDO	14.614***	14.920***	8.586	8.463
		(4.853)	(4.928)	(5.219)	(5.172)
Sponsor type variables	Small U.S. bank sponsor	4.211	4.754	18.481*	18.877*
		(6.013)	(6.279)	(10.116)	(10.036)
	Non-U.S. bank sponsor	−6.971	−7.426	−3.194	−3.058
		(6.048)	(6.081)	(4.897)	(4.908)
	Nonbanking sponsor	−3.875	−4.283	−2.031	−2.037
		(4.166)	(4.205)	(5.155)	(5.175)
Macro variables	Spread of one-month LIBOR over OIS	132.070		−9.411***	
		(189.638)		(2.431)	
	Volatility of the spread of one-month LIBOR over OIS	26.177		−17.917**	
		(107.050)		(7.259)	
	Return on the ABX index		−0.497		−0.873***
			(1.460)		(0.237)
	Volatility of the return on the ABX index		6.770		−1.826
			(4.737)		(1.316)
	Constant	−1.590	7.035	25.969***	20.749**

Table VII. Panel regressions of average maturity of new issues

6 Interpretive synthesis

The paper's evidence supports a three-part view of the 2007 ABCP collapse. First, the market suffered an aggregate shock tied to subprime mortgage risk and interbank funding stress. Second, investors mapped this aggregate uncertainty into program-level rollover decisions using public proxies for hidden exposure and support quality. Third, market outcomes differed by severity: some programs entered runs, while others still issued but at higher spreads and shorter maturities.

This synthesis links shadow banking to classic banking-panic theory. ABCP was not insured deposit funding, but it performed similar maturity transformation: short-term investors funded longer-term and opaque asset portfolios. Once investors doubted asset values and support strength, the market became vulnerable to runs. The key difference from traditional bank runs is that the run occurred in wholesale money markets and transmitted stress back to banks through liquidity commitments and implicit sponsor support.

7 Short summary

- ABCP outstanding fell about $350 billion in the final five months of 2007.
- The paper defines a program-level run as no issuance in a week with more than 10% of paper maturing, with the run continuing until issuance resumes.
- About one-third of programs experienced a run within weeks of the crisis onset, and nearly 40% were in a run by year-end.
- Runs were concentrated in programs with weaker liquidity support, lower ratings, riskier support providers, and riskier program types.
- Programs that still issued faced spreads and maturities that varied systematically with the same risk variables.
- LIBOR-OIS volatility and other macro-financial measures help explain the timing and severity of runs, spreads, and maturities.
- The evidence supports asymmetric-information theories of runs rather than a purely random panic.
- The paper shows how shadow-bank conduits can transmit stress back to banks through support commitments and balance-sheet exposure.

8 Conclusion

8.1 Core conclusion

Covitz, Liang, and Suarez show that the ABCP collapse of 2007 involved genuine program-level runs. These runs were not random: they were related to public indicators of program weakness, support quality, and macro-financial stress.

8.2 Economic intuition

ABCP investors were short-term creditors funding opaque asset pools. When subprime and money-market shocks raised uncertainty, investors could not perfectly identify which programs were exposed. They therefore used observable proxies, such as program type, rating, support structure, liquidity-provider strength, and sponsor type. Programs with weak signals either could not roll debt, paid higher spreads, or issued at shorter maturities.

8.3 Incremental contribution revisited

The paper advances the literature by measuring runs directly in a shadow-banking market, showing that the incidence of runs follows the logic of asymmetric-information bank-panic models, and documenting that price and maturity margins changed alongside quantity rationing. This makes the paper valuable for understanding why securitized short-term debt markets can be fragile even when formal bank deposits are insured.

8.4 Final takeaway

The ABCP crisis was a run on shadow-bank liabilities. Its pattern was both panic-like and information-based: the shock was aggregate, but withdrawal was selective. The paper's main lesson is that opacity plus short-term debt can make off-balance-sheet finance vulnerable to rapid market breakdown, with consequences for bank balance sheets and broader financial stability.